

Thrust 1

Measuring Phrasing Sensitive Failure

RQ1.1: How sensitive are medical VLMs to semantically equivalent paraphrases of clinical questions?

RQ1.2: How does explanation differ between correct and incorrect predictions, and when do we see Misleading Explanation Effect?

Thrust 2

Causal analysis of Failure Mechanisms

RQ2.1: Which architectural components or token pathways are causally responsible for phrasing sensitivity in VLMs?

RQ2.1: How do natural direct effects versus natural indirect effects contribute to prediction flips with stable visual focus?

Thrust 3

Mitigating Phrasing Sensitivity

RQ3.1: Can parameter-efficient fine-tuning reduce phrasing sensitivity while preserving diagnostic accuracy?

RQ3.2: How effective are consistency losses, contrastive alignment, and paraphrase augmentation for improving linguistic robustness?

Thrust 3

Mitigating Phrasing Sensitivity

RQ3.2: Will the safety finetuning and guard rail transfer from one modality to another?

RQ3.2: What residual risks remain under adaptive paraphrase-based attacks, and how should evaluation capture them?